

**Assignment #5\_A**

**SECTION A (2\*5=10 Marks)**

- Q1. What is intermediate code?
- Q2. What you mean by dead code Optimization?
- Q3. What is back patching?
- Q4. Explain in brief about control stack.
- Q5. Write down the benefits of using an intermediate code generation over direct code generation?

**SECTION B (7\*3=21 Marks)**

- Q1. Define runtime environment. What are the issues in runtime environment?
- Q2. Explain the Peephole Optimization in detail.
- Q3. Explain the DAG representation of Flow graph with suitable example.

**SECTION C (11\*3=33 Marks)**

- Q1.A.Explain the Storage allocation techniques in detail.
- Q1.B.Translate the following statement, which alters the flow of control of expressions, and generate the three-address code for it.

If (a<b) && (c<d) then t1 else t2.

- Q2.A.Explain the types of Three Address code with Example.
- Q2.B.Translate the following statement, which alters the flow of control of expressions, and generate the three-address code for it.

```
for (i=1;i<=n;i++)  
{  
  x=a+b*c;  
}
```

- Q3.A.Explain quadruples with the help of a suitable example.
- Q3.B.Translate the following statement, which alters the flow of control of expressions, and generate the three-address code for it.

```
Switch (i)  
{  
  Case 1:  
    x1=a1+b1*c1  
    Break;  
  Case 2:  
    x2=a2+b2*c2  
    Break;  
  Default:  
    x3=a3+b3*c3  
    Break;  
}
```